

1. Administrative & Study Identification

Full Study Title V114 and Acute Otitis Media (V114-032/PNEU-ERA)
Short Title/Acronym PNEU-ERA / V114-032 Protocol Number V114-032
/ 2023-001146-11 NCT Number NCT04193215 Sponsor Name Merck Sharp
& Dohme LLC Investigational Product V114 (15-valent pneumococcal
conjugate vaccine) Phase of Development Phase 3 Medical Monitor
Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * Primary Objective: To evaluate V114 in the prevention of vaccine-type acute otitis media (VT-AOM) and the safety of V114 with respect to the proportion of participants with serious adverse events (SAEs) through completion of the study. * Secondary Objectives: To evaluate the number of participants with VT-AOM and the percentage of participants who died from any cause during the study.

2.2 Background & Rationale To address the ongoing pediatric health burden caused by pneumococcal infections, specifically Acute Otitis Media (AOM). The primary hypothesis is that V114 (a 15-valent pneumococcal conjugate vaccine) is superior to no V114 in preventing vaccine-type acute otitis media as assessed by the incidence of VT-AOM.

3. Study Design & Methodology

3.1 Design Framework * Study Type: Interventional * Control Type: Active/Placebo ("No V114" arm) * Blinding: Double-blind / Observer-blind * Randomization: Randomized

3.2 Planned Enrollment & Duration * Target \$\$\$: Approximately 570 infants (based on related sample sizing) * Number of Sites: Multicenter (including sites in Thailand such as Chulalongkorn University, Siriraj Hospital, etc.) * Estimated Study Start: October 2020 * Estimated Primary Completion: December 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Is healthy (based on a review of medical history and physical examination). 2. Is male or female, approximately 2 months of age (from 42 days to 90 days of age).

4.2 Exclusion Criteria 1. Was born prior to 37 weeks of gestation. 2. Has a history of invasive pneumococcal disease (IPD) (positive blood culture, positive cerebrospinal fluid culture, or other sterile site) or known history of other culture-positive pneumococcal disease. 3. Has a known hypersensitivity to any component of the pneumococcal conjugate vaccine (PCV), any component of the licensed pediatric vaccines to be administered concomitantly in the study, or any diphtheria toxoid-containing vaccine. 4. Has any contraindication to the concomitant study vaccines being administered in the study, or has external auditory canal atresia/stenosis.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Scheduled pediatric immunization series * **Route of Administration:** Intramuscular (IM) injection * **Duration of Treatment:** Vaccinations and follow-up through completion of the study (typically up to 2-3 years of age)

5.2 Concomitant & Prohibited Therapy * **Permitted:** Concomitant licensed pediatric vaccines * **Prohibited:** Prior contraindicated PCV or diphtheria toxoid-containing vaccines

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Parent/Guardian Informed Consent, Medical History, Physical Exam
Baseline	Day 0	Randomization, First Vaccine Dose (IM Injection)
Interim	Scheduled Visits	Vaccine administration, safety monitoring, SAE evaluation, and VT-AOM clinical surveillance (visits at 4, 6, 12, 18, 24 months, etc.)
End of Study	Final Follow-up	Final SAE evaluation, concluding VT-AOM incidence assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Incidence of vaccine-type acute otitis media (VT-AOM) and the proportion of participants with serious adverse events (SAEs) through completion of the study. * **Measurement Method:** Clinical diagnosis, confirmation via culture testing, and systematic SAE tracking.

7.2 Secondary & Exploratory Endpoints * Number of participants with VT-AOM. * Percentage of participants who died from any cause.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Parent/caregiver diary, HCP interview during scheduled pediatric clinic visits.
 - **Serious Adverse Events (SAEs):** Defined as an AE that results in death, is life-threatening, requires or prolongs an existing hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is another important medical event deemed such by medical or scientific judgment.
 - **Data Monitoring Committee (DMC):** Internal/External safety review committee details per standard trial protocol.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy; Safety Set for all AE/SAE evaluations.
 - **Sample Size Justification:** Sample size scaled to adequately power the primary hypothesis that V114 is superior to no V114 in preventing VT-AOM.
 - **Handling of Missing Data:** Per protocol standards (e.g., Last Observation Carried Forward or Multiple Imputation for specific efficacy tracking).
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and remote monitoring of trial sites (e.g., vaccination documentation and AE reporting).
 - **Audit Trail:** Data captured via eCRF, tracking of AOM clinical cultures, and data clarification forms.
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11. Study Identifiers & Governance

Primary ID: V114-032 **Registry Number:** NCT04193215 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** V114 and Acute Otitis Media (V114-032/PNEU-ERA) **Official Regulatory Title:** V114 and Acute Otitis Media (V114-032/PNEU-ERA)

12. Clinical Framework (Study Specific)

Primary Condition: Acute Otitis Media (AOM) / Pneumococcal infection **Diagnosis Threshold:** Clinical presentation of AOM and culture-positive verification of VT pneumococcus **Medical Methodology:** Prophylactic immunization via pneumococcal conjugate

vaccine (V114) **Optimization Target:** Prevention of Vaccine-Type Acute Otitis Media (VT-AOM)

13. Study Procedures & Methodology

Management Pathway: Standard pediatric vaccination pathway and AOM surveillance protocol **Education Protocol (HCP):** Site investigator training on vaccine administration, proper AOM culture extraction, and SAE reporting criteria **Education Protocol (Patient):** Parent/caregiver training on identifying AOM symptoms and utilizing the safety reporting diary **Quality Audit Mechanism:** Investigator diagnosis tracking, laboratory culture validation, and caregiver diary adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: Up to completion of the pediatric study phase (approximately 2 to 3 years) **Onsite Visit Cadence:** Clinic visits scheduled every 2 months until 6 months of age, every 3 months until 1 year of age, and every 6 months until 2 years of age **Remote Monitoring Frequency:** Intermittent AE/SAE check-ins between scheduled visits **Checklist Review Interval:** Reviewed at each scheduled immunization and follow-up appointment

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				